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Assignment - 02

01) The probability of a pen manufactured by a factory

01) The joint probability distribution of discrete random variables X & Y are as follows

X/Y	-3	2	4
1	0.1	0.2	0.2
2	0.3	0.1	0.1

Then i) determine marginal distribution of X & Y ii) show that X & Y are dependent.

→

X/Y	-3	2	4	f(x)
1	0.1	0.2	0.2	0.5
3	0.3	0.1	0.1	0.5
g(y)	0.4	0.3	0.3	0.1

i) Marginal distribution of X

X	1	3
f(X)	0.5	0.5

Marginal distribution of Y

Y	-3	2	4
g(Y)	0.4	0.3	0.3

ii) $E(X) = \sum_{i=1}^{M} x_i f(x_i)$

$= 1 \times 0.5 + 3 \times 0.5$

$E(X) = 2$

$$E(Y) = \sum_{j=1}^{} y_j g(y_j)$$

$$= -3 \times 0.4 + 2 \times 0.3 + 4 \times 0.3$$

$$E(Y) = 0.6$$

ii) $$E(XY) = \sum_{i=1}^{} \sum_{j=1}^{} x_i y_j f_{ij}$$

$$= 1(-3)(0.1) + 1(2)(0.2) + 1(4)(0.2) + 3(0.3)(-3) + 3(2)(0.1) + 3(4)(0.1)$$

$$E(XY) = 0$$

$$E(X) \cdot E(Y) = 1.2$$

ii) $E(X)\,E(Y) \neq E(XY)$

So X & Y are dependent.

02) Determine the value of k so that the function $f(x,y) = k|x-y|$, for $x = -2, 0, 2$; $y = -2, 3$ represents joint probability dist of the random variable x and y. Also determine cov(X, Y).

→

X \ Y	-2	3	f(x)
-2	0	5k	5k
0	2k	3k	5k
2	4k	k	5k
g(x)	6k	9k	Total = 15k

By definition

$$\sum f(x_i, y_j) = 1$$

$$15k = 1$$

$$k = 1/15$$

Marginal distribution of x

x	-2	0	2
f(x)	5k	5k	5k

D D M M Y Y Y Y

Marginal distribution of Y

y	-2	3
g(y)	6k	9k

$$E(X) = \sum_{i=1}^{m} x_i f(x_i)$$

$$= -2 \times 1/3 + 0 + 2 \times 1/3$$

$$= 0$$

$$E(Y) = \sum_{j=1}^{n} y_j g(y_j)$$

$$= -2 \times 6/15 + 3 \times 3/5$$

$$= 1$$

$$E(XY) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j \bar{J}_{ij}$$

$$= -2(-2)0 + -2(3)5k + 0 + 0 + 2(-2)4k + 6k$$

$$= -40k$$

$$= -40/15$$

$$E(XY) = -8/3$$

$$Cov(X,Y) = E(X,Y) - E(X) \cdot E(Y)$$

$$= \frac{-8}{3} - 0$$

$$Cov(X,Y) = \frac{-8}{3}$$

03) Three boy
X always
ball to Z.
Y or a
to throw
ball aft
→ state s
TPM:

P =

D D M M Y Y Y Y

03) Three boys X, Y, Z are throwing a ball to each other. X always throws the ball to Y & Y always throws the ball to Z. But Z is just as likely to throw the ball to Y or as to X. Write TPM if Z is the first person to throw the ball, find the probability that X has the ball after fourth throw.

→ State space $\{X, Y, Z\}$

TPM:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix} \qquad P^{(0)} = (0, 0, 1)$$

$$S_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$P^2 = P \cdot P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

$$P^3 = P^2 \cdot P = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

$$P^4 = P^2 \cdot P^2 = \begin{pmatrix} 0 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{pmatrix}$$

$$S_4 = P^4 S_0 = \begin{pmatrix} 0 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$S_4 = \begin{pmatrix} 0.5 & 0.5 & 0.25 \end{pmatrix}$$

From hereby the probability that X has the ball after the forth throw is 0.5

D D M M Y Y Y Y

Q4) Given the following joint distribution of the random variable X & Y.

X/Y	-2	-1	4	5
1	0.1	0.2	0	0.3
2	0.2	0.1	0.1	0

Determine the marginal probability distributions of X & Y. Also compute Expectations of X, Y & XY, (ii) (iv) Cov(X,Y) (v) Correlation of X & Y

→

X/Y	-2	-1	4	5	f(x)
1	0.1	0.2	0	0.3	0.6
2	0.2	0.1	0.1	0	0.4
g(Y)	0.3	0.3	0.1	0.3	1

i) Marginal distribution of X

x	1	2
f(x)	0.6	0.4

Marginal distribution of Y

y	-2	-1	4	5
g(y)	0.3	0.3	0.1	0.3

ii) $E(X) = \sum_{i=1}^{n} x_i f(x_i)$

$= 1 \times 0.6 + 2 \times 0.4$

$E(X) = 1.4$

$E(Y) = \sum_{j=1}^{n} y_j g(y_j)$

$= -2(0.3) - 1(0.3) + 4(0.1) + 5(0.3)$

$E(Y) = 1$

$$E(XY) = \sum_{i=1}^{M} \sum_{j=1}^{N} x_i y_j \; ; \; i,j$$

$$= 1(-2)(0.1) + 1(-1)(0.2) + 1(0)(4) + 0.3(5)1 + 2(-2)(0.2) + 2(-1)(0.1) + 2(4)(0.1) + 2(5)(0)$$

$$E(XY) = 0.9$$

$$E(X)\,E(Y) = 1.4$$

$$E(XY) \neq E(X)\,E(Y)$$

$\therefore X$ & Y are dependent.

(ii) $Cov(X,Y) = E(XY) - E(X) \cdot E(Y)$

$= 0.9 - 1.4$

$Cov(X,Y) = -0.5$

$\sigma x = \sqrt{E(X^2) - (E(X))^2}$

$E(X^2) = \sum_{i=1}^{M} x_i^2 f(x_i)$

$= 1 \times 0.6 + 2^2 \times 0.4$

$E(X^2) = 2.2$

$\sigma x = \sqrt{2.2 - (1.4)^2}$

$\sigma x = 0.4899$

$\sigma y = \sqrt{E(Y^2) - (E(Y))^2}$

$E(Y^2) = \sum_{j=1}^{3} y^2 g(y_i)$

$= (-2)^2 0.3 + (-1)^2 0.3 + 16(0.1) + 25(0.3)$

$E(Y^2) = 10.6$

$\sigma y = \sqrt{10.6 - 1}$

$\sigma y = 3.0984$

(iv) $\rho(X,Y) = \dfrac{Cov(X,Y)}{\sigma x \cdot \sigma y} = \dfrac{-0.5}{0.4899 \times 3.0984}$

$\rho(X,Y) = -0.3294$

D D M M Y Y Y Y

a5) The joint probability distribution of random variables X & Y are as follows.

X/Y	-4	2	7
1	1/8	1/4	1/8
5	1/4	1/8	1/8

then determine i) Marginal distribution of X & Y ii) E(X), E(Y) & E(XY), iii) Cov(X, Y) iv) ρ(X, Y).

→

X/Y	-4	2	7	f(x)
1	1/8	1/4	1/8	1/2
5	1/4	1/8	1/8	1/2
g(y)	3/8	3/8	1/4	Total = 1

i) Marginal distribution of X

X	1	5
f(x)	1/2	1/2

Marginal distribution of Y

Y	-4	2	7
g(y)	3/8	3/8	1/4

ii) $E(X) = \sum_{i=1}^{n} x_i f(x_i)$

$= 1(1/2) + 5(1/2)$

$E(X) = 3$

$E(Y) = \sum_{j=1}^{n} y_j g(y_j)$

$= -4(3/8) + 2(3/8) + 7(1/4)$

$E(Y) = 1$

$$E(X,Y) = \sum_{i=1}^{M} \sum_{j=1} x_i y_j J_{ij}$$

$$= 1(-4)\,1/8 + 1(2)\,1/4 + 1(7)\,1/8 + 5(-4)\,1/4 + 5(2)\,1/8 + 35(1/8$$

$$E(X,Y) = 3/2$$

(iii) $COV(X,Y) = E(XY) - E(X) \cdot E(Y)$

$$= 3/2 - 3$$

$$COV(X,Y) = -3/2$$

(iii) $\sigma x = \sqrt{E(X^2) - (E(X))^2}$

$$E(X^2) = \sum_{i=1}^{M} x^2 f(x)$$

$$= 1^2(1/2) + 5^2(1/2)$$

$$E(X^2) = 13$$

$$\sigma x = \sqrt{13 - 3^2}$$

$$\sigma x = 2$$

$\sigma y = \sqrt{E(Y^2) - (E(Y))^2}$

$E(Y^2) = \sum_{j=1} y^2 g(x)$

$-(-4)^2 3/8 + 4(3/8) +$

$7^2(1/4)$

$E(Y^2) = 79/4$

$\sigma y = \sqrt{79/4 - 1}$

$\sigma y = 4.32$

(iv) $\rho(X,Y) = \dfrac{COV(X,Y)}{\sigma x \cdot \sigma y} = \dfrac{-3/2}{2 \times 4.33} = -0.1732$

$$\rho(X,Y) = -0.1732$$

06) The students study habits are as follows; If he studies on one night, he is 60% sure not study on next night. On the other hand, if he does not study on to night on next night. On the other hand, if he does not study on to night, he is 80% sure to study next night. Write the transition probability matrix for his chaen of study. In the long run how often does he study? Suppose he studies on Monday night, what is the probability that he

D D M M Y Y Y Y

does not study on Friday night?

→ Let A is studying B is not studying

state { A B }

$$P = \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix}$$

To find v:

$v = (x \ y)$ $\quad x+y=1$

$vp = v$ $\quad y = 1-x \quad \text{---} ①$

$$(x \ y) \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix} = (x \ y)$$

$$(0.4x+0.2y \ , \ 0.6x+0.8y) = (x \ y)$$

$$0.4x+0.2y = x$$

FROM ①

$$0.4x + 0.2(1-x) = x$$

$$0.4x + 0.2 - 0.2x = x$$

$$0.2x + 0.2 = x$$

$$0.2 = x - 0.2x$$

$$0.2 = 0.8x$$

$$\langle x = 0.25 \rangle$$

$$y = 1-x$$

$$\langle y = 0.75 \rangle$$

$v(x \ y) = (0.25, 0.75)$

(studying not studying)

07) Find the unique fixed probability vector of

$$P = \begin{pmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

→ Let $P = \begin{pmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ & v be the unique fixed probability vector.

$vP = v$

$v = (x\ y\ z)$; $x + y + z = 1$

$$(x\ y\ z)\begin{pmatrix} 0 & 3/4 & 1/4 \\ 1/2 & 1/2 & 0 \\ 0 & 1 & 0 \end{pmatrix} = (x\ y\ z)$$

$$\left(0x + \frac{y}{2} + 0z,\ \frac{3x}{4} + \frac{y}{2} + z,\ \frac{x}{4} + 0 + 0\right) = (x\ y\ z)$$

$\frac{y}{2} = x$ $\quad \frac{3x}{4} + \frac{y}{2} + z = y$ $\quad \frac{x}{4} = z$

$y = 2x$ $\quad x = 4z$

$z = \frac{x}{4}$

$x + y + z = 1$

$x + 2x + \frac{x}{4} = 1$

$3x + \frac{x}{4} = 1$ $\quad y = 2x$ $\quad z = x/4$

$13x = 4$ $\quad = 2 \times \frac{4}{13}$ $\quad = \frac{4}{4 \times 13}$

$\langle x = 4/13 \rangle$ $\quad \langle y = 8/13 \rangle$ $\quad \langle z = 1/13 \rangle$

08) Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for Santro. However, if he had a Santro, he is just as likely to trade it for a new Santro as to trade as to trade it for a Maruti or an Ambassador. In 2000 he bought his first car which was a Santro. Find the probability that he has

i) 2002 Santro
ii) 2002 Maruti
iii) 2003 Ambasador
iv) 2003 Santro

→ {Maruti ambassador santro}

{A B C}

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{pmatrix} \qquad P^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 4/9 & 4/9 \end{pmatrix}$$

$$P^3 = P \cdot P^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \end{pmatrix}$$

$$P^3 = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \\ 4/27 & 7/27 & 16/27 \end{pmatrix}$$

× [So in the long run, the probability that he] ×

Initially he has santro

$$P^{(0)} \cdot P^2 = (0,0,1) \begin{pmatrix} 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \\ 1/9 & 4/9 & 4/9 \end{pmatrix}$$

$$= (1/9, 4/9, 4/9)$$

i) P(he has santro in 2002) = 4/9
ii) P(he has Maruti in 2002) = 1/9
iii) P(he has Ambasador in 2003) = 7/27
iv) P(he has santro in 2003) = 16/27

9) Define probability vectors, stochastic matrices, Regular stochastic Matrix, stationary destribution and absorbing state of Markov chain.

→ Probability vector: A vector $B = B_1, B_2, \ldots B_n$ is called a probability vector.

stochastic Matrix: A square Matrix P having every row in the form of a probability vector is called stochastic Matrix.

* A stochastic Matrix is said to be regular stochastic Matrix if all the entries of the sum power p^n is positive
* The stationary destribution represents the long-term behaviour of the Markov chain.
* An absorbing state is a state that, once entered, cannot be left.

10) A salesman's territory consists of 3 cities, A, B and C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. However, if he sells in either B or C, then the next day he is twice as likely to sell in city A as in other city. In long run, how often does he sell in each of the cities

→ States {A, B, C}

TPM:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{pmatrix}$$

TO find V

$$V = (x, y, z) \quad ; \quad x+y+z = 1$$

$$VP = V$$

$$(x \; y \; z) \begin{pmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{pmatrix} = (x \; y \; z)$$

D D M M Y Y Y Y

$\left(\frac{2y}{3}+\frac{2z}{3}, \frac{xy}{3}+\frac{z}{3}, \frac{y}{3}\right) = (x\ y\ z)$

$z = \frac{y}{3}$ $\quad xy + \frac{z}{3} = y$

$x = \frac{2y}{3} + \frac{2z}{3}$ $\quad y + \frac{y}{9} = y$

$x = \frac{4yz}{3}$ $\quad 10y = 9y$

$x + y + z = 1$

$x + \frac{z}{3} = y \quad z = \frac{y}{3}$

$x + \frac{z}{3} = 3z \quad y = 3z$

$x = 3z - z/3$

$x = 8z/3$

$x + y + z = 1$

$\frac{8z}{3} + 3z + z = 1$

$8z/3 + 4z = 1$

$12z + 8z = 3$

$20z = 3$

$\langle z = 3/20 \rangle$

$y = 3 \times 3/20$

$\langle y = 9/20 \rangle$

$x = 8 \times \frac{3}{20 \times 3}$

$\langle x = 8/20 \rangle$

11) If the joint probability destribution of x and y is given by $f(x,y) = \frac{x+y}{30}$, for $x = 0,1,2,3$; $y = 0,1,2$

Find i) $P[x \leq 2, y = 1]$, ii) $P[x > 2, y \leq 2]$ iii) $P[x > y]$

x/y	0	1	2	g(x)
0	0	1/30	1/15	1/10
1	1/30	1/15	1/10	1/5
2	1/15	1/10	2/15	3/10.
3	1/10	2/15	1/6	2/5
g(y)	1/5	1/3	7/15	Total=1

i) $P(x \leq 2, y = 1)$

$P((x = 0, 1, 2), y = 1)$

$P(0,1) + P(1,1) + P(2,1)$

$\frac{1}{30} + \frac{1}{15} + \frac{1}{10}$

$= 1/5$

ii) $P(x \geq 2, y \leq 1)$

$P((x = 2, 3), y = 0, 1)$

$P(3,0) + P(3,1)$

$\frac{1}{10} + \frac{2}{15}$

$= 7/30$

iii) $P(x > y)$

$P(1,0) + P(2,0) + P(3,0) + P(2,1) + P(3,1) + P(3,2)$

$\frac{1}{30} + \frac{1}{15} + \frac{1}{10} + \frac{1}{10} + \frac{2}{15} + \frac{1}{6}$

$= 3/5$

12) Find the unique fixed probability vector for the regular stochastic matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$$

→

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix}$$

$VP = V \quad \therefore V = (x, y, z) \quad ; x + y + z = 1$

D D M M Y Y Y Y

$$(x \quad y \quad z)\begin{pmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix} = (x \quad y \quad z)$$

$$\left(\frac{y}{6}, x+\frac{y}{2}+\frac{2z}{3}, \frac{y}{3}+\frac{z}{3}\right) = (x \quad y \quad z)$$

$$\frac{y}{6} = x \qquad \frac{y}{3}+\frac{z}{3} = z$$

$$y = 6x \text{ —①} \qquad \frac{y}{3} = \frac{2z}{3}$$

$$y = 2z \text{ —②}$$

equate both eqn ① & ② $\quad z = y/2$

$$6x = 2z$$

$$3x = z \text{ —③}$$

$$x+y+z = 1 \qquad y = 6/10 \qquad \langle z = 3/10 \rangle$$

$$x+6x+3x = 1 \qquad \langle y = 3/5 \rangle$$

$$10x = 1$$

$$\langle x = 1/10 \rangle$$

13) Show that the matrix

$$\begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$ is a regular stochastic matrix

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/8 & 1/2 & 3/8 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$$

P^5 all the entries are positive.

∴ Given P is regular stochastic Matrix.

14) Three boys A, B & C are throwing a ball to each other. A is just as likely to throw the ball to B as C. B always throws the ball to A, and C is just as likely to throw the ball to A as to B. Find the probability that C has the ball after three throws if now A has the ball.

state space {A B C}

TPM :

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

$\text{Cov}(X,Y) = -8/3$

D D M M Y Y Y Y

$$P^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

$$P^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{pmatrix}$$

$P^3 = P \cdot P^2$

$$P^3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{pmatrix}$$

$$P^3 = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{pmatrix}$$

$\therefore$ C has the ball.

P(A has the ball) = 1/4

15) Given the following joint distribution of the random variables X and Y. Find the corresponding Marginal distribution and also compute the covariance.

X/Y	1	3	9
2	1/8	1/24	1/12
4	1/4	1/4	0
6	1/8	1/24	1/12

X/Y	1	3	9	f(x)
2	1/8	1/24	1/12	1/4
4	1/4	1/4	0	1/2
6	1/8	1/24	1/12	1/4
g(y)	1/2	1/3	1/6	total = 1

DD MM YYYY

Marginal distribution of x

x	2	4	6
f(x)	1/4	1/2	1/4

Marginal distribution of y

y	1	3	9
g(y)	1/2	1/3	1/6

$$E(X) = \sum_{i=1}^{M} x_i f(x_i)$$

$$= 2 \times \frac{1}{4} + 4 \times \frac{1}{2} + 6 \times \frac{1}{4}$$

$$E(X) = 4$$

$$E(Y) = \sum_{j=1}^{D} y_j g(y_j)$$

$$= 1 \times 1/2 + 3 \times 1/3 + 9 \times 1/6$$

$$E(Y) = 3$$

$$E(XY) = \sum_{i=1}^{M} \sum_{j=1}^{D} x_i y_j J_{ij}$$

$$= 2 \times 1(1/8) + 2(3)\,1/24 + 2(9)\,1/12 + 4(1)\,1/4 + 4(3)\,1/4$$
$$+ 4(9)\,0 + 6(1)\,1/8 + 6(3)\,1/24 + 6(9)\,1/12$$

$$E(XY) = 12$$

$$COV(XY) = E(XY) - E(X) \cdot E(Y)$$

$$= 12 - 4 \times 3$$

$$COV(XY) = 0$$

D D M M Y Y Y Y

16) Show that $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is a regular stochastic matrix. Also find the associated unique fixed probability vector.

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix} \qquad P^3 = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix} \qquad P^5 = \begin{bmatrix} 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/8 & 3/8 & 1/2 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/8 & 3/8 & 1/2 \end{bmatrix}$$

P^5 all entries are positive.

$\therefore$ Given P is regular stochastic matrix.

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

$VP = V \quad ; V = (x, y, z) \qquad x+y+z = 1$

$$(x \ y \ z) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} = (x \ y \ z)$$

$$\left(\frac{z}{2}, x + \frac{z}{2}, y\right) = (x \ y \ z)$$

$$\frac{z}{2} = x \qquad x + \frac{z}{2} = y \qquad y = z$$

$x+y+z = 1 \qquad \langle x = 1/5 \rangle$

$\frac{z}{2} + z + z = 1 \qquad \langle y = 2/5 \rangle$

$5z = 2$

$\langle z = 2/5 \rangle$

D D M M Y Y Y Y

(17) The joint probability distribution of two discrete random variables X and Y is $f(x,y) = k(2x+y)$, where x and y are integers such that $0 \le x \le 2$, $0 \le y \le 3$.

i) Find the value of constant k.

ii) Show that the random variables X and Y are dependent.

iii) Find $P(X \ge 1, Y \le 2)$

$f(x,y) = k(2x+y)$ $\quad x = 0,1,2 \quad y = 0,1,2,3$

x/y	0	1	2	3	f(x)
0	0	k	2k	3k	6k
1	2k	3k	4k	5k	14k
2	4k	5k	6k	7k	22k
g(y)	6k	9k	12k	15k	Total = 42k

i) By definition

$\Sigma f(x_i, y_j) = 1$

$42k = 1$

$k = 1/42$

ii) $E(X) = \sum_{i=1}^{m} x_i f(x_i)$

$= 0 + 1(14k) + 2(22k)$

$= 58k \Rightarrow 58k$

$= 58/42 \quad 29/21$

$E(Y) = \sum_{j=1}^{n} y_j g(y_j)$

$= 0 + 1(9k) + 2(12k) + 3(15k)$

$= 78k \quad 78k$

$E(Y) = 78/42 \quad 13/7$

$E(XY) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j \cdot J_{ij}$

$= 0 + 1(k)0 + 1(3k) + 2(4k) + 3(5k) + 0 + 2(5k) + 4(6k) + 6(7k) = 102k$

$= 95k \Rightarrow 95/42 \quad 102/42 \Rightarrow 17/7$

$E(XY) = E(X) \cdot E(Y) \quad E(XY \neq E(X) \cdot E(Y)$

$\frac{17}{7} \neq \frac{377}{147}$ $\therefore$ Given X & Y are dependent.

$\therefore E(X,Y) = \quad$ $\text{Cov}(X,Y) = E(XY) \quad = -8/3 - 0$

$V(X,Y) = -8/3$

D D M M Y Y Y Y

18) A fair coin is tossed thrice. The random variables X & Y are defined as X=0 or 1 according as head or tail occurs on the first toss, Y-number of heads. Compute $\text{Cov}(X,Y)$

→ A coin is tossed 3 times, the sample space S consists of $2^3 = 8$.

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

Each outcome has $p(\text{outcome}) = 1/8$

The random variable X = 0 or 1

X=1 if the 1st toss is head(H)

$\{HHH, HHT, HTH, HTT\}$

X=0 if the 1st toss is a tail(T)

$\{THH, THT, TTH, TTT\}$

probability distribution of X is

$p(X=0) = 4/8 = 1/2 \qquad p(X=1) = 4/8 = 1/2$

random variable Y is the no. of heads in three tosses

$Y=0 \{TTT\} \quad Y=1 \{HTT, THT, TTH\}$

$Y=2 \{HHT, HTH, THH\} \quad Y=3 \{HHH\}$

M.D of X

x	0	1
f(x)	1/2	1/2

M.D of Y

y	0	1	2	3
g(y)	1/8	3/8	3/8	1/8

$E(X) = \sum_{i=1}^{M} x_i f(x_i) = 0 + 1/2 = 1/2$

$E(Y) = \sum_{j=1}^{n} y_j g(y_j) = 0 + 3/8 + 2(3/8) + 3/8 = 3/2$

$E(XY) = \sum_{i=1}^{} \sum_{j=1}^{} x_i y_j f_{ij} = 3(1/8) + 2(1/8) + 2(1/8) + 1/8 + 0 = 1$

$\text{Cov}(XY) = E(XY) - E(X) \cdot E(Y)$

$= 1 - 3/2\,(1/2)$

$\therefore \text{Cov}(XY) = 1/4$